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RESEARCH ARTICLE

Phase Current Equalization Method of IPMSM-Based on-Board Boost Converter for Electric Vehicles

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ABSTRACT As the distance traveled per charge increases, the adoption of 800V batteries is increasing. However, most electric vehicle (EV) charging stations are designed to charge 400V batteries. However, if the installed motor and inverter operate as a boost converter, the 800V battery can be charged using an existing 400V charger. As a result of this increased availability, on-board boost charging is becoming more common. When an internal permanent magnet synchronous machine (IPMSM) is used as the boost inductor, the shape of the three-phase current changes because the current ripple varies with the rotor angle due to the magnetic saliency of the rotor. In addition, if the averages of the phase currents are not equal, torque can be produced. In this study, the complete IPMSM inductance model is utilized to obtain an analytical representation of the current ripple as a function of the switching duty and rotor angle. Based on this, the maximum charging current is estimated and the maximum charging power is also derived as a function of the rotor angle. It also determines the optimal rotor angle that minimizes the current ripple. This study also addresses phase current equalization to avoid torque generation. To this end, a compensation method is proposed to obtain an accurate average current in the double-sampling method, which samples the top and valley. The derived formulas are verified via simulations and experiments.

INDEX TERMS Integrated on-board charger, boost converter, IPMSM, EV, current equalization, average current sampling.

I. INTRODUCTION

As the driving distance per charge increases, the battery capacity tends to increase. However, the charging time is limited, so the power capacity of the charger must be increased. By the way, the charger ampere rating cannot be increased due to weight limitations on the charger cables and plugs. Therefore, to increase the charging power, there is no choice but to increase battery voltage. As a result, electric vehicles equipped with 800V batteries appear in the market and are gaining popularity.

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Currently, 800V chargers are not as widespread as 400V chargers [1], [2]. The convenience of charging 800V electric vehicle (EV) will be greatly expanded if the vehicle has a built-in boost converter, so that a 400V charger can charge an 800V battery [1]. Based on this idea, some EVs have been built to charge 800V batteries using both 400V and 800V chargers [3]. In these EVs, the motor and inverter can serve as inductor and boost circuits, respectively. Integrated chargers utilizing propulsion motor and inverter have been extensively studied for a long time. Apart from DC charging, AC integrated chargers were primarily studied in the early days [4], [5]. Eull et al. [4] focused on the common-mode voltage and leakage current in a grid-connected integrated charger

and reduced the bearing currents by adding an LC filter and common-mode inductor. Pando-Acedo et al. [5] proposed a method for reducing the pulsating torque in a single-phase integrated charger by regulating the q -axis current to zero depending on the rotor angle. Shi et al. [6] split motor phase coils: one as a PFC inductor for a single-phase grid and the other two as boost inductors. A similar approach was adopted in [7], but a current-source rectifier was added before the internal permanent magnet synchronous machine (IPMSM).

Charging methods using multiple motors and inverters have also been investigated [8], [9], [10], [11]. Sul and Lee [8] proposed a charging method for four-wheel EV using four motors and four inverters. Two of them were connected to the grid-PFC, whereas the other two were used as buck-type choppers for charging the battery. Woo et al. [9] provided inductance guidelines for the current ripple to be within an acceptable range while the motor is used for the PFC. They added an additional boost inductor between the grid and the motor neutral point to further reduce the current ripple [10].

Multi-phase motors, which have more than three phases, are used as charger inductors for AC grid to reduce torque ripple and switch current stress [12], [13], [14], [15], [16], [17], [18]. Various types of motors have been investigated for integrated charging applications, such as permanent magnet assisted synchronous reluctance motors [19], switched reluctance motors [20], [21], [22], [23], [24], interior permanent magnet synchronous motors with damper bars [25], and open-end winding motors [26], [27], [28].

A multi-phase charger with multiple inductors is often used to increase the power rating. It also has the advantage of reducing current ripple via interleaving. A group of researchers investigated the pros and cons of coupled inductors in multi-phase system [29], [30], [31], [32], [33], [34]. In coupled inductors, the magnetic core is shared by multiple coils so that the phase currents affect each other through the magnetic flux linkage. There are two types of coupling: direct and inverse, which depend on the sign of the mutual inductance. Inversely coupled inductors have received more attention because they can reduce the current ripple and core size compared to uncoupled inductors in some interleaved conditions [29], [30], [31], [32], [33], [34]. From the perspective of the charger inductor, the IPMSM looks like an inversely coupled three-phase coil, with a coupling coefficient of $-1/2$ [35].

In multi-phase chargers, it is common to use interleaved PWM, but in the case of motor-based integrated chargers, a synchronous PWM method that applies the same switching signal is used for fear of torque generation [36], [37], [38]. Xiao et al. [38] added an additional inductor to the neutral point of the motor to mitigate the current ripple in synchronous PWM. However, synchronous PWM is not desirable from a current ripple point of view. Pellegrino et al. [39] applied interleaved PWM to an integrated charger. They controlled the dq components to zero on average to suppress

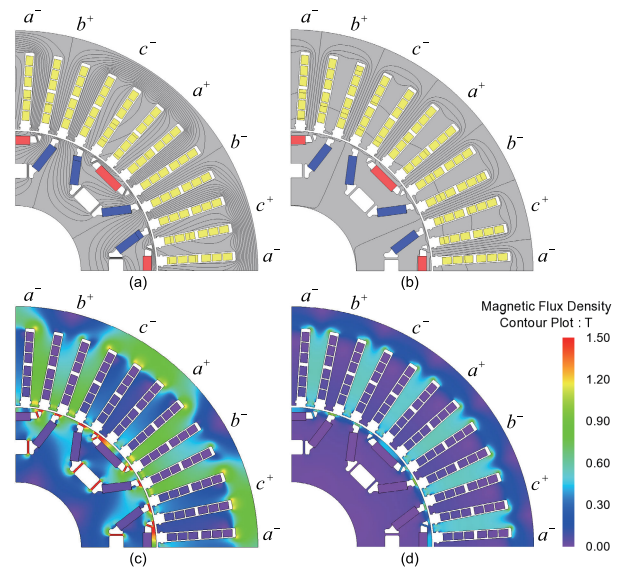


FIGURE 1. Flux lines and density at different frequencies; (a), (c) 100Hz sinusoidal current and (b), (d) 10kHz triangle wave current.

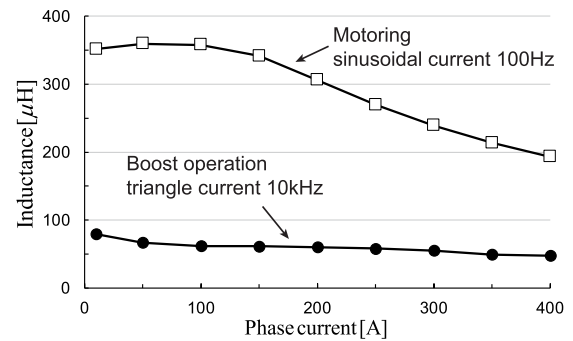


FIGURE 2. Inductance variation of a motor phase coil depending on frequency and current shape: motoring with 100Hz sinusoidal current and boost operation with 10kHz triangle wave current (FEM simulation).

torque generation. However, because they did not include rotor saliency in their analysis, they could not account for the current ripple variation with rotor angle and phase current inequality.

In this study, a complete analysis of an integrated charger is pursued based on the full inductance model of the IPMSM. The proposed method uses interleaved PWMs of equal duty that move 120° from each other. The phase currents are controlled to have the same average current to make the shaft torque zero on average. The analysis includes the current ripple variation depending on the rotor angle caused by the rotor magnetic saliency. Based on the ripple magnitude, the feasible average current is calculated within the switch current limit. Correspondingly, the maximum charging power is obtained as a function of the rotor angle, and the optimum angles are found that maximizes the charging power while minimizing the current ripple. In addition, a compensation method is developed to correct the average of the

TABLE 1. Parameters of IPMSM.

Parameter	Value	Parameter	Value
Max. power	15 kW	Rated torque	90 Nm
Nom. DC voltage	72 V	Base speed	1600 rpm
Max. speed	8000 rpm	Max. current (rms)	330 A
Mutual ind. (L_m)	125 μ H @100 Hz 50 μ H @10 kHz	Coil-resistance. (r_s)	0.009 Ω
Leakage ind. (L_{ls})	45 μ H @100 Hz 75 μ H @10 kHz	No. pole pair (p)	4 pairs
Saliency ind. (L_δ)	30 μ H @100 Hz 40 μ H @10 kHz		

double-sampling to the actual current average for currents with many vertices.

This paper is organized as follows. The charging scheme is presented along with the inductance model of IPMSM in Section II. The peak-to-peak value of the current ripple is derived, as well as the average, as a function of switching duty and rotor angle. Also, the maximum charge power is estimated in Section III. A current control method with a compensation term to compensate for the current sampling error is described in Section IV.

II. OPERATION PRINCIPLE

A. IPMSM AS A CHARGER INDUCTOR

Since the IPMSM is a rotating machine, the core and winding configuration is not optimized for use as a charger inductor. In particular, the rotor has permanent magnets for torque generation and many narrow holes to reduce torque ripple and cogging torque. Furthermore, to generate reluctance torque, the magnetic circuit of the rotor is designed to be anisotropic. These are not only useless, but also harmful when charging. Owing to this magnetic saliency, the coupling coefficient between the coils changes according to the rotation angle of the rotor. In addition, in the case of distributed winding, the mutual inductance is relatively large; therefore, the ripple current tends to increase when charging with zero-phase current components.

IPMSMs are designed to operate in the relatively low-frequency region 0 ~ 1 kHz whereas the charger operating frequency is 10 ~ 20 kHz. This results in a large difference in inductance when the IPMSM is used as a motor versus the charger inductor. These features are illustrated with FEM simulation results for a 15 kW, 72 V IPMSM used in this experiment. Its major parameters are listed in Table 1. Fig. 1 shows the flux lines and density of the IPMSM for different applications. The remaining two results are obtained when it is under motoring with 100 Hz, 100 A (rms) sinusoidal current. The eddy current effect was included in the simulation for 0.35 mm laminated steel sheet. The right two are the results when 10 kHz boost operation takes place with 100 A average current. Note that the flux line does not penetrate deep inside the rotor at high frequencies. This can also be verified with the flux densities shown in Fig. 1 (c) and (d).

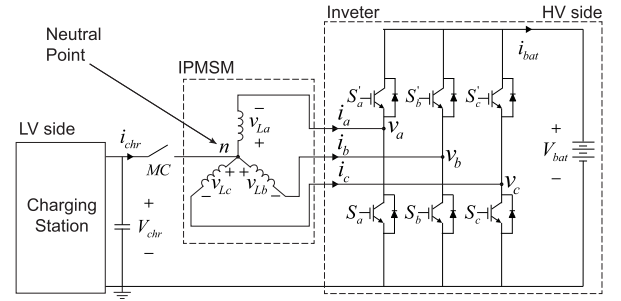
**FIGURE 3.** Configuration of the integrated boost charger.

Fig. 2 shows the phase coil inductance at different frequencies. When motoring with 100 Hz sinusoidal current, the inductance is much greater than when charging with 10 kHz switching. This is because much of the high frequency field is hindered from passing into the rotor core owing to the eddy current effect. This causes the inductance to appear low. On the other hand, the leakage inductance increases as the field leaks around the stator teeth and rotor surface. Therefore, it is necessary to look IPMSM inductance differently in high-frequency areas. For example, in motoring, the inductances are $L_m = 125 \mu$ H, $L_{ls} = 45 \mu$ H, $L_\delta = 35 \mu$ H, while $L_m = 50 \mu$ H, $L_{ls} = 75 \mu$ H and $L_\delta = 40 \mu$ H at 10 kHz boost charging.

B. CHARGER CONFIGURATION

Fig. 3 shows a typical integrated charger where the inverter is used as the boost circuit and the IPMSM is used as the boost inductor. It also shows that the motor coil is wound with a wye winding and the neutral point is connected to the positive rail of the DC charger through magnetic contact switch. It is assumed that the on-board battery V_{bat} is higher than the charger voltage V_{chr} . For example, $V_{chr} = 400$ V and $V_{bat} = 800$ V.

Since the IPMSM has magnetic saliency on the rotor, the inductance matrix can be divided into fixed and angle-dependent parts [35]:

$$\bar{\mathbf{L}}_{abc}(\theta) \triangleq \begin{bmatrix} L_{aa} & L_{ab} & L_{ac} \\ L_{ba} & L_{bb} & L_{bc} \\ L_{ca} & L_{cb} & L_{cc} \end{bmatrix} = \mathbf{L}_{abc} - \mathbf{L}_{rlc}(\theta), \quad (1)$$

where

$$\mathbf{L}_{abc} = \begin{bmatrix} L_m + L_{ls} & -L_m/2 & -L_m/2 \\ -L_m/2 & L_m + L_{ls} & -L_m/2 \\ -L_m/2 & -L_m/2 & L_m + L_{ls} \end{bmatrix}, \quad (2)$$

$$\mathbf{L}_{rlc}(\theta)$$

$$= L_\delta \begin{bmatrix} \cos 2p\theta & \cos(2p\theta - \frac{2\pi}{3}) & \cos(2p\theta + \frac{2\pi}{3}) \\ \cos(2p\theta - \frac{2\pi}{3}) & \cos(2p\theta + \frac{2\pi}{3}) & \cos 2p\theta \\ \cos(2p\theta + \frac{2\pi}{3}) & \cos 2p\theta & \cos(2p\theta - \frac{2\pi}{3}) \end{bmatrix}, \quad (3)$$

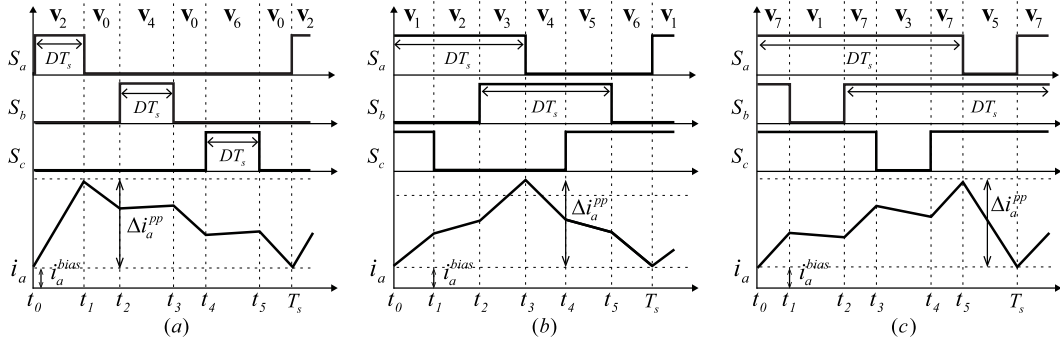


FIGURE 4. Interleaved PWM sequences and the corresponding α -phase current; (a) Mode 1: $[0 \leq D < 1/3]$, (b) Mode 2: $[1/3 \leq D < 2/3]$, (c) Mode 3: $[2/3 \leq D < 1]$.

L_m is the mutual inductance, L_{ls} is the stator winding leakage inductance, L_δ is a saliency constant determined by the geometry of the PM cavities, θ is the mechanical rotor angle, and p is the pole pair of the IPMSM. Let the inverter pole voltages be denoted by $\mathbf{v}_{abc} = [v_a, v_b, v_c]^T$, the stator phase currents by $\mathbf{i}_{abc} = [i_a, i_b, i_c]^T$, and the phase voltage by $\mathbf{v}_L = [v_{La}, v_{Lb}, v_{Lc}]^T \equiv [V_{chr} - v_a, V_{chr} - v_b, V_{chr} - v_c]^T$. The voltage equation of the IPMSM is [35]

$$\mathbf{v}_L = r_s \mathbf{i}_{abc} + \frac{d}{dt} \left\{ \bar{\mathbf{L}}_{abc}(\theta) \mathbf{i}_{abc} + \boldsymbol{\psi}(\theta) \right\}, \quad (4)$$

where the PM flux linkage $\boldsymbol{\psi}(\theta)$ is

$$\boldsymbol{\psi}(\theta) = \psi_m \begin{bmatrix} \cos p\theta \\ \cos(p\theta - 2\pi/3) \\ \cos(p\theta + 2\pi/3) \end{bmatrix}, \quad (5)$$

and ψ_m is the back electromotive force (EMF) constant. The coil resistance r_s is usually ignored in converter analysis and is therefore not included in the following analysis. The rotor does not rotate, so that $\frac{d\theta}{dt} = 0$. Then it follows that

$$\frac{d}{dt} \mathbf{i}_{abc} = \bar{\mathbf{L}}_{abc}^{-1}(\theta) \mathbf{v}_L(t). \quad (6)$$

However, it is difficult to obtain $\bar{\mathbf{L}}_{abc}^{-1}(\theta) = [\mathbf{L}_{abc} - \mathbf{L}_{rlc}(\theta)]^{-1}$ manually in a closed form. With the help of MATLAB symbolic manipulation, we get the inverse matrix as (7), shown at the bottom of the next page, where $\alpha = (3L_m^2 - 3L_\delta^2)/L_{ls} + 2L_m$ and $\alpha^+ = 3\alpha + 4L_{ls} + 6L_m$.

C. CHARGING SCHEME

The following analysis is based on an interleaved PWM scheme in which the switching duty is the same for all phases and they are shifted 120° . As shown in Fig. 4, the interleaved PWM is divided into three modes of operation according to duty D , i.e., Mode 1: $[0 \leq D < 1/3]$, Mode 2: $[1/3 \leq D < 2/3]$, and Mode 3: $[2/3 \leq D < 1]$. The on-duties do not overlap between phases in Mode 1, but they do overlap in Mode 2 and 3. Depending on the switching state, the phase

voltage is separated into eight states:

$$\mathbf{v}_L = \begin{cases} \mathbf{v}_1 = [V_{chr}, V_{chr} - V_{bat}, V_{chr}]^T \\ \mathbf{v}_2 = [V_{chr}, V_{chr} - V_{bat}, V_{chr} - V_{bat}]^T \\ \mathbf{v}_3 = [V_{chr}, V_{chr}, V_{chr} - V_{bat}]^T \\ \mathbf{v}_4 = [V_{chr} - V_{bat}, V_{chr}, V_{chr} - V_{bat}]^T \\ \mathbf{v}_5 = [V_{chr} - V_{bat}, V_{chr}, V_{chr}]^T \\ \mathbf{v}_6 = [V_{chr} - V_{bat}, V_{chr} - V_{bat}, V_{chr}]^T \\ \mathbf{v}_7 = [V_{chr} - V_{bat}, V_{chr} - V_{bat}, V_{chr} - V_{bat}]^T \\ \mathbf{v}_0 = [V_{chr}, V_{chr}, V_{chr}]^T. \end{cases} \quad (8)$$

Note that $\mathbf{v}_0$ and $\mathbf{v}_7$ represent the states of common mode voltage. Specifically, only Mode 1 has $\mathbf{v}_0$'s, while Mode 3 only has $\mathbf{v}_7$'s. There is no common mode voltage in Mode 2. Fig. 5 shows the phase voltages and currents in Mode 2 in which the current slope depends on the voltage state.

Substituting (8) and (7) into (6), the α -phase current slope equation is obtained in each sub-interval such that

$$\frac{di_a}{dt} = \begin{cases} \frac{V_{chr}\alpha^+ - V_{bat}(\alpha + 4L_\delta \cos(2p\theta - \frac{2\pi}{3}))}{L_{ls}\alpha^+} \equiv k_{a1}, \\ \frac{V_{chr}\alpha^+ - V_{bat}(2\alpha - 4L_\delta \cos(2p\theta))}{L_{ls}\alpha^+} \equiv k_{a2}, \\ \frac{V_{chr}\alpha^+ - V_{bat}(\alpha + 4L_\delta \cos(2p\theta + \frac{2\pi}{3}))}{L_{ls}\alpha^+} \equiv k_{a3}, \\ \frac{V_{chr}\alpha^+ - V_{bat}(\alpha^+ - \alpha - 4L_\delta \cos(2p\theta - \frac{2\pi}{3}))}{L_{ls}\alpha^+} \equiv k_{a4}, \\ \frac{V_{chr}\alpha^+ - V_{bat}(\alpha^+ - \alpha + 4L_\delta \cos(2p\theta))}{L_{ls}\alpha^+} \equiv k_{a5}, \\ \frac{V_{chr}\alpha^+ - V_{bat}(\alpha^+ - \alpha - 4L_\delta \cos(2p\theta + \frac{2\pi}{3}))}{L_{ls}\alpha^+} \equiv k_{a6}, \\ \frac{V_{chr} - V_{bat}}{L_{ls}} \equiv k_{a7}, \\ \frac{V_{chr}}{L_{ls}} \equiv k_{a0}. \end{cases} \quad (9)$$

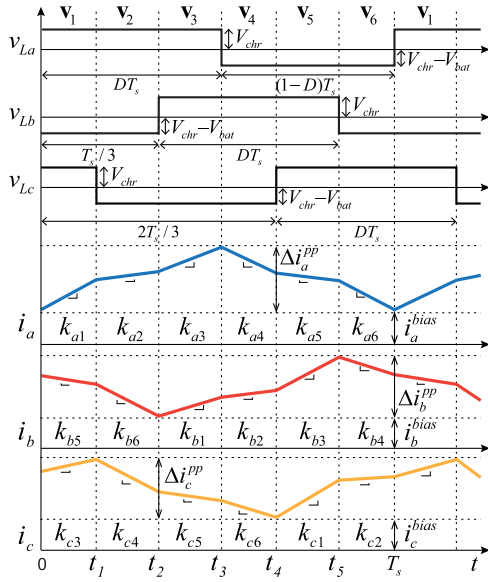


FIGURE 5. Phase voltages and currents in Mode 2.

For convenience of notation, the current slope of each section is marked with $k_0, \dots$, and k_7 . The inductor currents of b and c phases are essentially the same, but different with respect to the rotor angle θ . Specifically, the current slopes of b and c phases are shifted from those of a -phase by 120° :

$$k_{bn}(\theta) = k_{an}(\theta + \frac{\pi}{3p}), \quad n = 1, 2, \dots, 6, \quad (10)$$

$$k_{cn}(\theta) = k_{an}(\theta - \frac{\pi}{3p}), \quad n = 1, 2, \dots, 6. \quad (11)$$

Although the slope differs by 120° between the phases, the phase currents are not congruent, as shown in Fig. 5.

D. STEADY STATE VOLTAGE RATIO

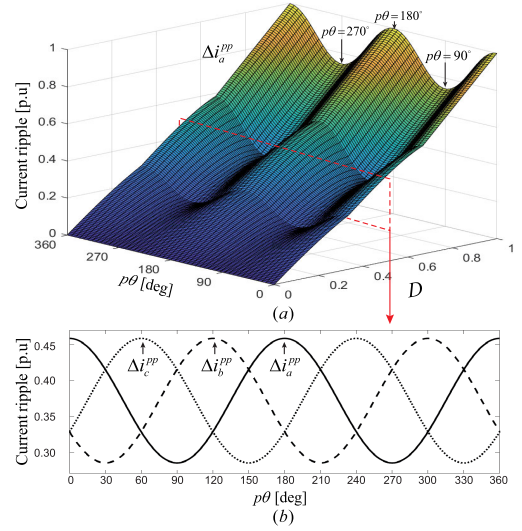
The voltage ratio between V_{chr} and V_{bat} can be determined as a function of D in steady state. In this section, we determine the voltage ratio between V_{chr} and V_{bat} as function of D at steady state. To do this, the zero voltage-time law is first applied to the inductor in Mode 2 as follows.

$$i_a(T_s) - i_a(0) = \int_0^{T_s} \frac{di_a(t)}{dt} dt = \sum_{n=1}^6 k_{an} \delta i_{an} = 0, \quad (12)$$

where

$$\delta i_{a1} = k_{a1}(t_1 - 0) = k_{a1}(D - \frac{1}{3})T_s$$

$$\delta i_{a2} = k_{a2}(t_2 - t_1) = k_{a2}(\frac{2}{3} - D)T_s$$

FIGURE 6. (a) α -phase current ripple versus switching duty D and rotor angle $p\theta$, (b) ripple magnitude variations of phase coil with respect to rotor angle $p\theta$ when $D = 0.5$.

$$\delta i_{a3} = k_{a3}(t_3 - t_2) = k_{a3}(D - \frac{1}{3})T_s$$

$$\delta i_{a4} = k_{a4}(t_4 - t_3) = k_{a4}(\frac{2}{3} - D)T_s$$

$$\delta i_{a5} = k_{a5}(t_5 - t_4) = k_{a5}(D - \frac{1}{3})T_s$$

$$\delta i_{a6} = k_{a6}(t_6 - t_5) = k_{a6}(\frac{2}{3} - D)T_s \quad (13)$$

Substituting (9) into (13), it follows from (12) that

$$V_{bat} = \frac{1}{(1-D)} V_{chr}. \quad (14)$$

It is very surprising that the angular dependence on the voltage ratio has completely disappeared here. The resulting expression is equivalent to the voltage ratio of a conventional boost converter. Applying the zero voltage-time law on b and c gives the same result, although the currents are coupled in the inductor. Also, the same voltage ratio is obtained for Mode 1 and Mode 3, which can be easily verified using MATLAB symbolic manipulation.

III. CHARGER POWER CALCULATION

The phase current can be divided into the DC and ripple components. That is, $i_x = i_x^{bias} + \Delta i_x$, where i_x^{bias} is the DC bias component and Δi_x is the ripple component.

A. PEAK-TO-PEAK VALUE OF CURRENT RIPPLE

Let the peak-to-peak value of Δi_x be denoted by Δi_x^{pp} . As shown in Fig. 4, the ripple patterns are different depending

$$\bar{\mathbf{L}}_{abc}^{-1}(\theta) = \frac{1}{L_s \alpha^+} \begin{bmatrix} \alpha^+ - 2\alpha + 4L_s \cos(2p\theta) & \alpha + 4L_s \cos(2p\theta - \frac{2\pi}{3}) & \alpha + 4L_s \cos(2p\theta + \frac{2\pi}{3}) \\ \alpha + 4L_s \cos(2p\theta - \frac{2\pi}{3}) & \alpha^+ - 2\alpha + 4L_s \cos(2p\theta + \frac{2\pi}{3}) & \alpha + 4L_s \cos(2p\theta) \\ \alpha + 4L_s \cos(2p\theta + \frac{2\pi}{3}) & \alpha + 4L_s \cos(2p\theta) & \alpha^+ - 2\alpha + 4L_s \cos(2p\theta - \frac{2\pi}{3}) \end{bmatrix} \quad (7)$$

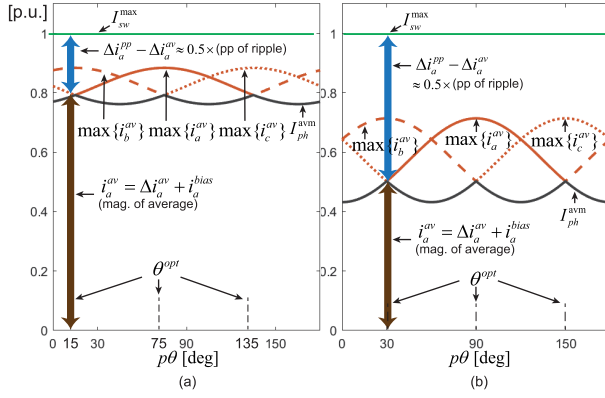


FIGURE 7. Relative magnitude of ripple and average current according to switch current limit i_{sw}^{max} (a) when $D = 0.2$, (b) when $D = 0.5$.

on the duty. Thereby, Δi_x^{pp} should be calculated differently depending on the modes.

1) MODE 1: $[0 \leq D < 1/3]$

Note from Fig. 4 (a) that the peak to peak value is determined in the first interval by the highest slope resulted from v_2 . Therefore, the peak to peak value is obtained from (9) such that

$$\begin{aligned} \Delta i_a^{pp}(\theta) &= k_{a2}DT_s \\ &= \frac{DV_{chr}T_s}{L_s\alpha^+} \left[\alpha^+ - \frac{(2\alpha - 4L_\delta \cos 2p\theta)}{(1-D)} \right], \end{aligned} \quad (15)$$

2) MODE 2: $[1/3 \leq D < 2/3]$

As shown in Fig. 4 (b), the peak to peak value consists three voltage components, v_1 , v_2 , and v_3 . So it is a function of k_{a1} , k_{a2} and k_{a3} defined in (9):

$$\begin{aligned} \Delta i_a^{pp}(\theta) &= T_s \left[k_{a1} \left(D - \frac{1}{3} \right) + k_{a2} \left(\frac{2}{3} - D \right) + k_{a3} \left(D - \frac{1}{3} \right) \right] \\ &= \frac{DV_{chr}T_s}{L_s\alpha^+} \left[\alpha^+ - \frac{(2\alpha - 4L_\delta \cos 2p\theta)}{3D(1-D)} \right], \end{aligned} \quad (16)$$

3) MODE 3: $[2/3 \leq D < 1]$

As shown in Fig. 4 (c), the peak to peak value is determined in the last interval with v_5 , such that

$$\begin{aligned} \Delta i_a^{pp}(\theta) &= -k_{a5}(1-D)T_s \\ &= \frac{DV_{chr}T_s}{L_s\alpha^+} \left[\alpha^+ - \frac{(2\alpha - 4L_\delta \cos 2p\theta)}{D} \right]. \end{aligned} \quad (17)$$

The current ripples of b and c phases are the same in shape, but differ in the phase of the rotor angle such that

$$\Delta i_b^{pp}(\theta) = \Delta i_a^{pp}(\theta + \frac{\pi}{3p}), \quad (18)$$

$$\Delta i_c^{pp}(\theta) = \Delta i_a^{pp}(\theta - \frac{\pi}{3p}). \quad (19)$$

The normalized plot of Δi_a^{pp} is shown as a 3-D plot of D and θ in Fig. 6 (a). The current ripple reaches its maximum, $\frac{V_{chr}T_s}{L_s\alpha^+}(\alpha^+ - 2\alpha + 4L_\delta)$ when $D = 1.0$ and $\theta = 0^\circ$. A section view at $D = 0.5$ is shown in Fig. 6 (b), in which the peak

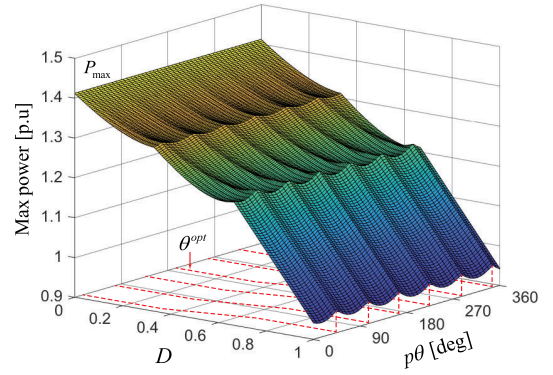


FIGURE 8. Maximum power versus D and $p\theta$ under switch current limit i_{sw}^{max} .

to peak values change as the rotor angle changes. Note that the ripples become maximum at $0^\circ, 60^\circ, 120^\circ, 180^\circ, \dots$ for a fixed D .

B. CURRENT AVERAGE

The current average consists of DC bias and the ripple average as follows:

$$i_x^{av} = i_x^{bias} + \Delta i_x^{av}, \quad (20)$$

where i_x^{av} is the average of i_x and $\Delta i_x^{av} \equiv \frac{1}{T_s} \int_0^{T_s} \Delta i_x dt$. Based on Fig. 5,

$$\begin{aligned} \Delta i_a^{av} &= \frac{1}{2T_s} \left[k_{a1}t_1^2 \right. \\ &\quad + \{2i_a(t_1) + k_{a2}(t_2 - t_1)\}(t_2 - t_1) \\ &\quad + \{2i_a(t_2) + k_{a3}(t_3 - t_2)\}(t_3 - t_2) \\ &\quad + \{2i_a(t_3) + k_{a4}(t_4 - t_3)\}(t_4 - t_3) \\ &\quad + \{2i_a(t_4) + k_{a5}(t_5 - t_4)\}(t_5 - t_4) \\ &\quad \left. + i_a(t_5)(T_s - t_5) \right]. \end{aligned} \quad (21)$$

After a long computation with the help of MATLAB symbolic manipulation, the ripple average is obtained as

$$\Delta i_a^{av}(\theta) = \begin{cases} \frac{DV_{chr}T_s}{2L_s\alpha^+} \left[\alpha^+ - \frac{2\alpha - \frac{8}{\sqrt{3}}L_\delta \cos(2p\theta - \frac{\pi}{6})}{(1-D)} \right] & \text{for } 0 \leq D < \frac{1}{3}, \\ \frac{DV_{chr}T_s}{2L_s\alpha^+} \left[\alpha^+ - \frac{2\alpha - 8\gamma L_\delta \cos(2p\theta - \beta)}{3D(1-D)} \right] & \text{for } \frac{1}{3} \leq D < \frac{2}{3}, \\ \frac{DV_{chr}T_s}{2L_s\alpha^+} \left[\alpha^+ - \frac{2\alpha - \frac{8}{\sqrt{3}}L_\delta \cos(2p\theta + \frac{\pi}{6})}{D} \right] & \text{for } \frac{2}{3} \leq D < 1. \end{cases} \quad (22)$$

where $\gamma = \sqrt{3D^2 - 3D + 1}$ and $\beta = \tan^{-1}(\sqrt{3}(1 - 2D))$. Similarly to (18) and (19), the ripple averages of b and c phases are obtained as shifted functions of θ :

$$\Delta i_b^{av}(\theta) = \Delta i_a^{av}(\theta + \frac{\pi}{3p}), \quad (23)$$

TABLE 2. Optimal rotor angles over D .

D	θ^{opt}
Mode 1 : $[0 \leq D < 1/3]$	$\frac{(4n-3)\pi}{12p}, n = 1, 2, \dots, 6p$
Mode 2 : $[1/3 \leq D < 2/3]$	$\frac{(2n-1)\pi}{6p} - \beta, n = 1, 2, \dots, 6p$
Mode 3 : $[2/3 \leq D < 1]$	$\frac{(4n-1)\pi}{12p}, n = 1, 2, \dots, 6p$

$$\beta = \tan^{-1}(\sqrt{3}(1-2D)).$$

$$\Delta i_c^{av}(\theta) = \Delta i_a^{av}(\theta - \frac{\pi}{3p}). \quad (24)$$

C. MAXIMUM CHARGING POWER

The maximum power is determined by the maximum charging current when the battery voltage is fixed. And it is determined by the semiconductor switch's current limit, I_{sw}^{max} , considering the margin, i.e.,

$$\max\{i_x\} = \Delta i_x^{pp}(\theta) + i_x^{bias} \leq I_{sw}^{max}. \quad (25)$$

If the current ripple is large, even if the average is low, the peak current can reach the maximum limit of the switch. However, such cases are undesirable. Conversely, it is desirable to reduce the ripple to maximize the fraction of the average current within the current limit.

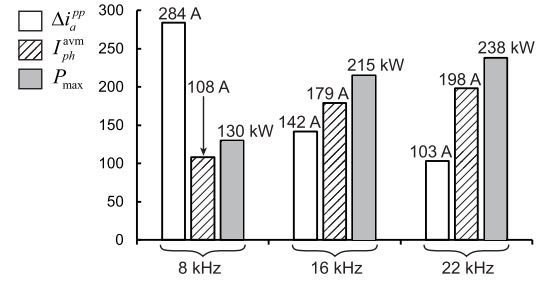
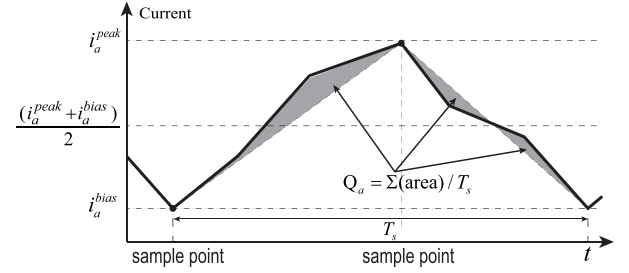
Substituting (20) into (25), it follows that

$$i_x^{av}(\theta) \leq I_{sw}^{max} + \Delta i_x^{av}(\theta) - \Delta i_x^{pp}(\theta). \quad (26)$$

In the case of a -phase, the average i_a^{av} can be expanded until the equality holds, i.e., the maximum average current is obtained when the equality holds:

$$\max\{i_a^{av}\}(\theta) \equiv \begin{cases} I_{sw}^{max} - \frac{DV_{chr}T_s}{2L_{ls}\alpha^+} [\alpha^+ - \frac{2\alpha - \frac{8}{\sqrt{3}}L_\delta \cos(2p\theta + \frac{\pi}{6})}{1-D}] & \text{for } 0 \leq D < \frac{1}{3}, \\ I_{sw}^{max} - \frac{DV_{chr}T_s}{2L_{ls}\alpha^+} [\alpha^+ - \frac{2\alpha - 8\gamma L_\delta \cos(2p\theta - \beta)}{3D(1-D)}] & \text{for } \frac{1}{3} \leq D < \frac{2}{3}, \\ I_{sw}^{max} - \frac{DV_{chr}T_s}{2L_{ls}\alpha^+} [\alpha^+ - \frac{2\alpha - \frac{8}{\sqrt{3}}L_\delta \cos(2p\theta - \frac{\pi}{6})}{D}] & \text{for } \frac{2}{3} \leq D < 1. \end{cases} \quad (27)$$

Fig. 7 shows plots of the maximum average currents, $\max\{i_a^{av}\}(\theta)$, $\max\{i_b^{av}\}(\theta)$, and $\max\{i_c^{av}\}(\theta)$ which depend on the rotor angle θ . Note that the plots are normalized by I_{sw}^{max} . This shows that the maxima of the averages are not the same, and they change differently with θ . Of course, they are also functions of switching duty D . Fig. 7 shows two cases for $D = 0.2$ and $D = 0.5$. The magnitude of the ripple increases with D , the average portion decrease as D increases. Only the average current is considered when calculating the maximum power. One strategy for this charging scheme is to equalize the charge power between the phases. Therefore, the lowest

**FIGURE 9.** Ripple magnitude, maximum average current, and maximum charging power depending on switching frequency when $I_{sw}^{max} = 250A$.**FIGURE 10.** Double-sampling and its average error denoted by shaded triangles.

value has to be selected in estimating the charger capacity. That is, the maximum average current for charging is

$$I_{ph}^{avm}(\theta) \equiv \min \left\{ \max\{i_a^{av}\}, \max\{i_b^{av}\}, \max\{i_c^{av}\} \right\}. \quad (28)$$

This is represented by bold black lines in Fig. 7. Thus, the maximum charging power is estimated as

$$P_{max} = 3V_{chr}I_{ph}^{avm}(\theta). \quad (29)$$

Fig. 8 shows a 3-D plot of P_{max} over D and θ . It is normalized by $3V_{chr}I_{sw}^{max}/\sqrt{2}$. Several optimal rotor angles θ^{opt} are found that maximize I_{ph}^{avm} . Table 2 shows the optimal angles. However, it is a function of D in Mode 2, where $1/3 < D < 2/3$, and the D -dependence is found via β .

Fig. 9 shows some calculation results when 800V battery is charged from a 400V charger using a fictitious IPMSM that has the same inductances shown in Table 1. The switch current limit is assumed to be 250A. It shows the ripple magnitude, Δi_a^{pp} , average current, I_{ph}^{avm} , and power, P_{max} based on the power estimation (29). Note that the charging power increases with the switching frequency as the ripple magnitude decreases dramatically.

IV. CURRENT CONTROL

As mentioned above, IPMSMs are different from dedicated boost inductors, because the phase coils are partially coupled and the degree of coupling is affected by the rotor angle. Fig. 11 shows the entire current control block diagram in which the phase currents are controlled separately but with interleaving.

During charger operation, the phase currents must be equal to avoid torque build-up. Therefore, the objective of this

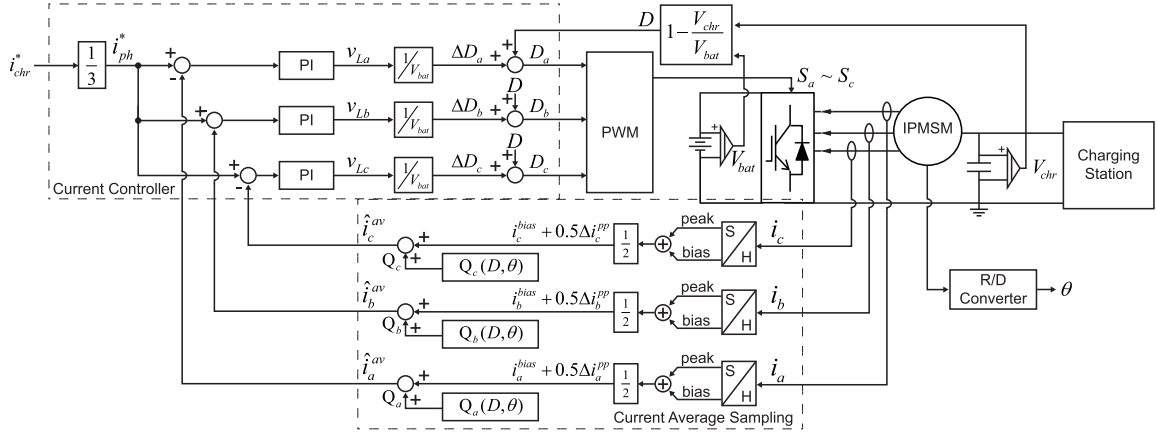


FIGURE 11. Overall current control block diagram that has the average compensation terms, $Q_a(D, \theta)$, $Q_b(D, \theta)$, and $Q_c(D, \theta)$.

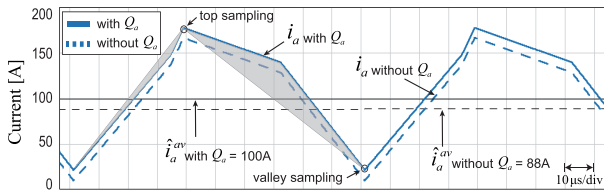


FIGURE 12. Simulation evidence showing a difference in phase current sampling with or without compensator Q_a .

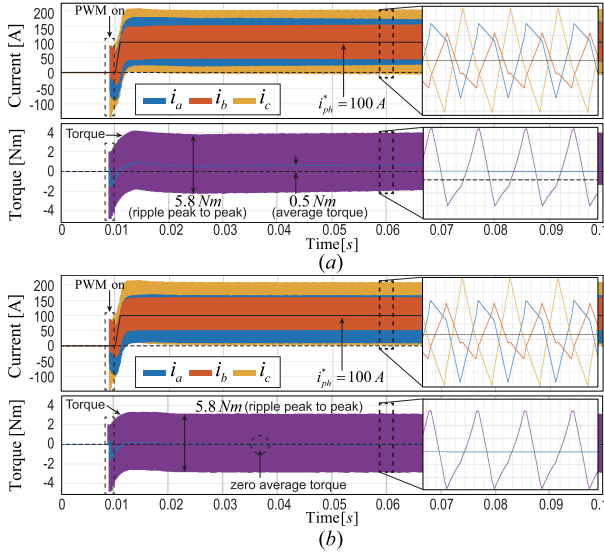


FIGURE 13. Effect of current average compensator Q_x : (a) phase currents and torque without the compensator, (b) with Q_a , Q_b , and Q_c .

controller is to equalize the average currents of all the phases at any rotor angle. Recall that the phase currents have different shapes owing to asymmetry in the rotor flux path. As a result, the plain double-sampling method does not provide an accurate average value, which samples the current at the top and valley corners. As shown in Fig. 4, the phase current exhibits multiple corner points that depend on the switching duty and the rotor angle.

Fig. 10 shows the sample points and graphical meaning of Q_a . The currents are sampled at the top and valley according

to the double-sampling. Note that $0.5(i_a^{peak} + i_a^{bias})$ is the average value. The shaded area indicates a possible sampling error when the average is calculated from the double-sampling data. The shaded area above the line indicates a ‘negative’ error, whereas that below the line indicates a ‘positive’ error. Therefore, the double-sampling result should be corrected by adding the area above the line and subtracting the area below the line. Now, we define the compensation term Q_x as the signed sum of shaded triangles, $Q_x = \sum \pm \{\text{shaded triangle area}\} / T_s$, and is obtained from (15)-(17) and (22) such that

$$Q_a(\theta) = \Delta i_x^{av}(\theta) - 0.5 \Delta i_a^{pp}(\theta)$$

$$= \begin{cases} \frac{2V_{chr}T_sL_\delta D}{\sqrt{3}L_{ls}\alpha^+(1-D)} \sin 2p\theta, & 0 \leq D < \frac{1}{3}, \\ \frac{2V_{chr}T_sL_\delta(1-2D)}{\sqrt{3}L_{ls}\alpha^+(1-D)} \sin 2p\theta, & \frac{1}{3} \leq D < \frac{2}{3}, \\ -\frac{2V_{chr}T_sL_\delta}{\sqrt{3}L_{ls}\alpha^+} \sin 2p\theta. & \frac{2}{3} \leq D < 1. \end{cases} \quad (30)$$

Likewise, the compensation terms on b and c phases are obtained as a function shifted with respect to θ :

$$Q_b(\theta) = Q_a(\theta + \frac{\pi}{3p}) \quad (31)$$

$$Q_c(\theta) = Q_a(\theta - \frac{\pi}{3p}) \quad (32)$$

The average current i_a^{av} can be rearranged from (20) and (30) such that

$$i_a^{av} = 0.5 \Delta i_a^{pp} + i_a^{bias} + Q_a$$

$$= 0.5(i_a^{peak} + i_a^{bias}) + Q_a. \quad (33)$$

Fig. 11 shows the overall current controller in which three PI controllers are applied for three phase currents and the sampling compensation terms $Q_a(\theta)$, $Q_b(\theta)$, and $Q_c(\theta)$ are added. The target charging current is divided into three for each phase; $i_{ph}^* = \frac{1}{3} i_{in}^*$. The steady state switching duty D set by (14) is equally added to the feedforward path.

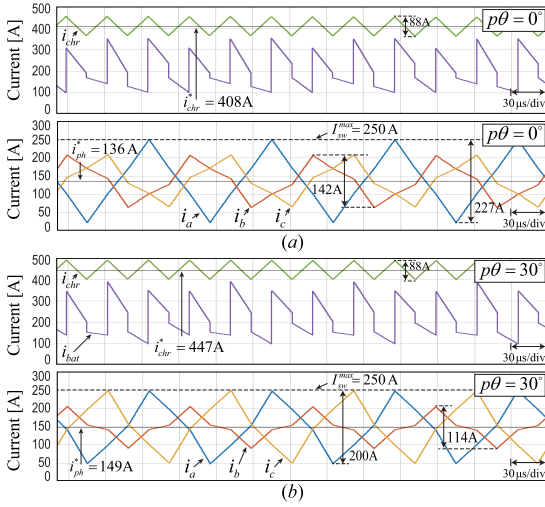


FIGURE 14. Charger current i_{chr} , battery current i_{bat} , and phase (inductor) currents i_a , i_b , and i_c for different angles: (a) $\theta = 0^\circ$ and (b) $\theta = 30^\circ$.

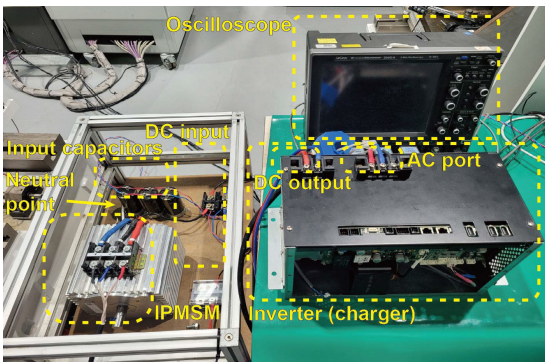


FIGURE 15. Experimental setup.

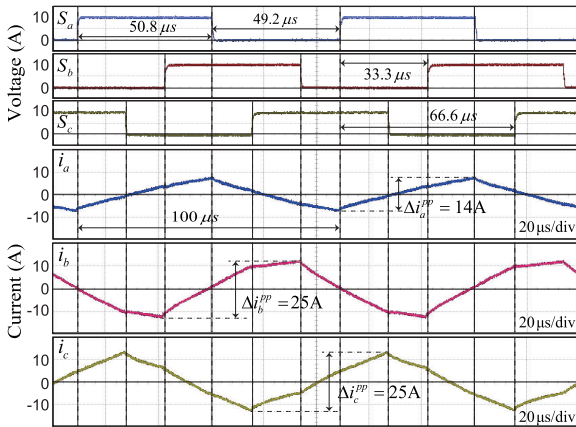


FIGURE 16. Switching sequence and phase currents when $p\theta = 90^\circ$.

V. SIMULATION AND EXPERIMENTAL RESULTS

The current simulation was performed using MATLAB Simulink with $V_{chr} = 400$ V, $V_{bat} = 800$ V, and $f_{sw} = 10$ kHz. The IPMSM parameters are listed in Table 1 were used. Figs. 12 and 13 show the effect of the average corrector Q_x on the current sampling. First, Fig. 12 shows the actual current i_a and the sampled average $\hat{i}_a^{av}$ when compensator Q_a is on and off. Shown as a solid line when Q_a is on, and as a dotted line otherwise. If Q_a is not present, the average

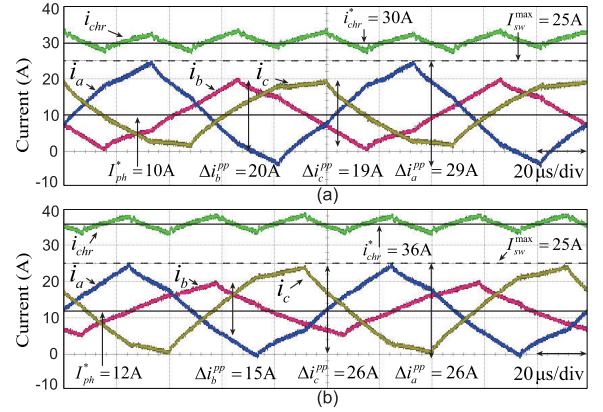


FIGURE 17. Input current and phase currents for different angles when $I_{sw}^{max} = 25$ A (a) $p\theta = 0^\circ$ (b) $p\theta = 30^\circ$.

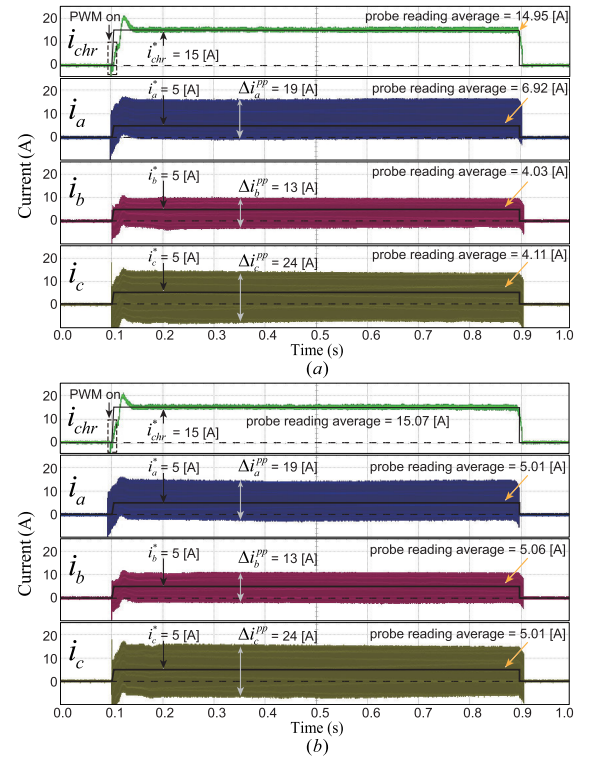


FIGURE 18. Step responses to input (charger) current command $i_{chr}^* = 15$ A when $p\theta = 45^\circ$: (a) without Q_x , (b) with Q_x .

value will appear lower than the true value because the shaded area is not considered. However, when Q_a , which reflects the shaded area, is added, $\hat{i}_a^{av}$ is adjusted to the actual average value. Fig. 13 shows the phase currents and torque with and without Q_x . Fig. 13 (b) clearly shows that 0.5 Nm torque in Fig. 13 (a) is nullified with the use of Q_x , which provides correct average currents for feedback.

Fig. 14 shows the simulation results of the charger current i_{chr} , battery current i_{bat} , and the phase (inductor) currents i_a , i_b , and i_c for different angles when $I_{sw}^{max} = 250$ A, $V_{chr} = 400$ V, and $V_{bat} = 800$ V. The switch current limit is assumed to be 250 A. Note from Table 2 that when $D = 0.5$, 30° is an optimal angle at which the average phase current (149 A) is maximized and the peak-to-peak value of the

ripple is minimized. The optimality of 30° is demonstrated by comparing the average current (136 A) at 0° . As a result, the battery charge current or/and power is 11% greater at $p\theta = 30^\circ$ than at $p\theta = 0^\circ$. A corresponding graphical illustration is shown in Fig. 7 (b).

Experiments were performed on a reduced scale with a motor, whose parameters are listed in Table.1. Fig. 15 shows a photo of the experimental setup. A 50 V DC power supply was used as a charger source, $V_{chr} = 50V$ and an electric load as a battery $V_{bat} = 100V$. The control board was made with a 32-bit DSP, TMS320F28388D and an inverter stack with an IGBT module. Fig. 16 shows the IGBT gate voltages of S_a , S_b , and S_c and the corresponding phase currents when $p\theta = 90^\circ$ and $D = 0.5$. Since there was no load, the current average is zero. The ripple magnitudes are calculated as $\Delta i_a^{pp} = 14.2$ A and $\Delta i_b^{pp} = \Delta i_c^{pp} = 24.8$ A according to (15)–(19) at $p\theta = 90^\circ$. Similar results are obtained, as shown in Fig. 16. Fig. 17 shows the difference in the maximum charging current depending on the rotor angle, where the switch current maximum was set to $I_{sw}^{max} = 25A$. The same on-duty cycle and rotor angle were set in the experiment, as in the simulation in Fig. 14. Similar to the case of the simulation, it is shown in Fig. 17 (a) and (b) that the maximum peak-to-peak ripple is reduced by 11% compared to the case of $p\theta = 0^\circ$ at the optimal angle, $p\theta = 30^\circ$. As a consequence, the average charging current can be increased by about 17% at the optimal angle. Fig. 18 shows the effect of the average compensator Q_x when a 15 A step command was applied at $V_{chr} = 60$ V, $V_{bat} = 100$ V, and $p\theta = 45^\circ$. In Fig. 18 (a), the average currents read by the current probes were not the same because the current control was performed without the average compensator. Meanwhile, in Fig. 18 (b), the same experiment was conducted with Q_x turned on. Observe that all the probe current readings are close to 5 A. It shows the effectiveness of Q_x in phase current equalization. The effect is greater in high voltage systems.

VI. CONCLUSION

Unlike a dedicated inductor, the IPMSM has a large leakage inductance due to the air gap, and the mutual inductance changes according to the rotor angle. Therefore, even if a voltage of the same duty is applied, the shape and average value of the ripple are different depending on the phase, and the degree of the difference varies depending on the position of the rotor. To overcome this complexity, a voltage equation was established based on the full inductance model of the IPMSM and the ripple current equation when the interleaved PWM voltage was applied was derived. The maximum charging power was calculated using the peak to peak value and average value of the current ripple.

Since the ripple has more edges in the interleaved PWM, average current cannot be obtained by sampling twice in one cycle. In this study, a compensation equation that corrects the error of the double-sampling method was derived. By using this compensation formula, the average current measurement is accurate, and proper current control using this feedback

can equalize the phase currents. This can suppress torque at any fixed rotor angle. This was verified by simulations and experiments.

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